

Colophon

Text and Graphics Layout

The layout of the book is unusual. There are two criteria that override everything else. First, a block of text discussing a graphical display needs to be as close as possible to the display, preferably on the same page or on a facing page. Second, it is vital to make content the master of design and not vice versa. Some approach text-display layout by writing to the design. This typically interferes with the coherence of the discussion. Throwing in a thought because space needs to be filled is not a good way to communicate. The design here solves the layout problem by giving up on filling and balancing pages, a convention thought to be inviolate by some. The spacing between paragraphs and the ragged-right text is meant to lessen the sense of visual order so that the lack of balance and fill is less salient.

Visualization Software Credits

The workhorse software of the book is S, the powerful system developed by Richard Becker, John Chambers, and Allan Wilks. The original visualizations of data that formed the basis of the writing were done in S. The only excursions out of S in the data analysis were to Geomview, the well designed software of Stuart Levy, Tamara Munzner, and Mark Phillips for direct manipulation 3-D graphics.

The final rendering of the displays for publication, however, used a number of systems. Except for the 3-D plots and the color level plot, all final rendering was done by proprietary software of MEM Research, Inc., written by Robert McGill. The software, which is based on S, enables very delicate control of the details of graphs so that graphical elements — such as labels, tick marks, and the positioning of the panels of multi-panel displays — can be placed precisely where they best serve

the visualization. The placement was carried out by Marylyn McGill of MEM with enormous attention to detail. The final product was PostScript[®] files.

The color level plot and the 3-D plot with shading traveled through a maze of software systems. Both started out in S as trivariate data sets. The color level plot was drawn by S and then moved to Adobe Photoshop for the assignment of CMYK colors to the levels. The information about the 3-D plot went to Geomview where rendering assignments were made — the lighting, the surface characteristics, and the orientation. Then it moved on to Renderman[®] to improve the quality of the image, and finally, went to Quark XPress[®] where the labels were added.

Production

This book was typeset in L^AT_EX. The main text is set in palatino, figure legends in helvetica, italics in times, and mathematical symbols in computer modern. Camera-ready copy was produced at AT&T Bell Laboratories on a Linotronic 200 P with a resolution of 1270 dpi.

Edwards Brothers, Inc., of Ann Arbor, Michigan, U.S.A., printed the book. The paper is 70 pound Sterling Satin, the pages are Smythe sewn, and the book is covered with Arrestox[®] linen.